Why does the kid swing alone in the park? Parametric resonance

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Why does the kid swing alone in the park? A child's first experiences on a swing often involve a parent pushing it to sustain its oscillations. Eventually, the child learns to sustain the oscillations by periodically moving their legs and changing their body position. This phenomenon is an example of parametric resonance, in which periodic variations of a system's parameters can amplify its oscillations. In this case, energy is transferred from the child's body, through the movement of their legs, to the mechanical energy of the swing.

Another remarkable example is the Botafumeiro, a large swinging censer used in the Cathedral of Santiago de Compostela since the Middle Ages to spread incense. Its oscillations can be amplified through parametric excitation; see the accompanying video.

Parametric resonance also appears in celestial mechanics. An important model is the restricted three-body problem, in which two massive bodies influence the motion of a third body whose mass is assumed to be negligible. For example, one may consider a satellite orbiting Earth in a plane inclined relative to the Earth–Sun orbital plane.

This project introduces parametric resonance using techniques from ordinary differential equations and geometric mechanics.

